Name: ____________ Student ID: ____________ Grade: ____________

DSAA 1001 Week 1 Validation Set

Probability, Distributions, and Transformations

Instructions

- Purpose: Self-assessment and practice before test
- Time: 30 minutes | Total: 120 pts
- Open book, AI tools allowed
- References: Learning Sheet 1, Sections 1-8

Part A: Multiple Choice [40 points]

Question 1. (5 pts) Which object is an event in a probability space?

a) a single numerical value of a random variable
b) a subset of the sample space
c) the mean of a distribution
d) a likelihood function viewed without a model

Answer: ______

Question 2. (5 pts) Let $\Omega = \{r, s, t\}$ with probabilities $0.25, 0.50, 0.25$. If $X(r) = 2$, $X(s) = 2$, and $X(t) = 5$, what is $P(X = 2)$?

a) 0.25
b) 0.50
c) 0.75
d) 1

Answer: ______

Question 3. (5 pts) A binomial PMF with parameters (n, p) relies essentially on which assumption?

a) The trials have independent outcomes with a common success probability
b) The number of trials is itself Gaussian
c) The success probability changes after every trial
d) All possible counts are equally likely

Answer: ______

Question 4. (5 pts) A screening test has a rare condition as its cause. Which statement best describes the base-rate issue in a posterior calculation?

a) A large likelihood always makes the posterior large
b) The prior prevalence still affects the posterior after the evidence is observed
c) The evidence term can be omitted when sensitivity is known
d) Conditioning makes the prior irrelevant

Answer: ______

Question 5. (5 pts) If a continuous variable has CDF F, which expression gives the probability of an interval $a < X \le b$?

a) $F(a) - F(b)$
b) $F(b) - F(a)$
c) $f(b) - f(a)$
d) $F(a)F(b)$

Answer: ______

Question 6. (5 pts) Let $I = \mathbf{1}_A$ be an indicator. Which identity is always true?

a) $E[I] = P(A)$
b) $E[I] = P(A)^2$
c) I must be independent of every variable
d) $\mathrm{Var}(I) = P(A)$

Answer: ______

Question 7. (5 pts) If $X \sim \mathcal{N}(-2, 1)$ and $Z = -3X + 4$, what is the variance of Z?

a) -3
b) 1
c) 3
d) 9

Answer: ______

Question 8. (5 pts) If $U \sim \text{Uniform}(0, 1)$ and $X = F^{-1}(U)$ for a continuous CDF F, what is the purpose of the construction?

a) It produces a sample with CDF F
b) It makes every value of X equally likely
c) It removes the need to define a CDF
d) It works only for Gaussian distributions

Answer: ______

Part B: Medium Question 9 [20 points]

Question 9. (20 pts) A monitoring model records the joint PMF of status S and alert A:

	$A = 0$	$A = 1$
S = normal	0.45	0.05
S = fault	0.15	0.35

(a) [4 marks] Give the marginal PMF of A.

(b) [6 marks] Compute $P(S = \text{fault} \mid A = 1)$ and explain in words what population this conditional probability refers to.

(c) [6 marks] Test whether S and A are independent using one joint-probability comparison.

(d) [4 marks] A modeller proposes a coarse sample space with only the outcomes {normal, fault}. Explain one question about alerts that this coarse model cannot answer.

Part C: Medium Question 10 [20 points]

Question 10. (20 pts) Let R have density $f_R(r) = 2(1 - r)$ for $0 \le r \le 1$ and 0 otherwise. A service has 12 independent-looking but not necessarily independent requests; request i is rejected with event B_i, with $P(B_i) = 0.08$.

(a) [6 marks] Derive the CDF $F_R(r)$ on $[0,1]$ and compute $P(R > 0.5)$.

(b) [8 marks] Let $M = \sum_{i=1}^{12} \mathbf{1}_{B_i}$. Compute $E[M]$ and explain why the calculation remains valid even if the rejection events are dependent.

(c) [6 marks] Suppose a two-group measurement model has $Y|G = 0 \sim \mathcal{N}(10, 1)$ and $Y|G = 1 \sim \mathcal{N}(20, 4)$. Explain why the mixture is not generally determined by its overall mean and variance alone, and give one feature that a single Gaussian can hide.

Part D: Hard Question 11 [40 points]

Question 11. (40 pts) Let X have density $f_X(x) = 3x^2$ for $0 < x < 1$. Define a folded score $Y = \begin{cases} x & 0 \leq x \leq \frac{1}{2} \\ 1-x & \frac{1}{2} < x \leq 1 \end{cases}$ when $X = x$.

(a) [14 marks] Derive $F_Y(y) = P(Y \leq y)$ by identifying all X-values that map into the event. Give the full piecewise CDF.

(b) [10 marks] Differentiate to obtain $f_Y(y)$ on the interior of the support and verify that it integrates to one. Explain how the two branches contribute different amounts of mass.

(c) [8 marks] Since $F_X(x) = x^3$ on $[0,1]$, describe inverse-transform sampling for X. For $U = 0.125$, compute X and the resulting Y.

(d) [8 marks] State two checks that would distinguish a correct folded-score implementation from an implementation that accidentally keeps only the $X \leq \frac{1}{2}$ branch. Explain why checks based on finite simulation are not proofs.

End of Validation Set — Check answers with AI and discuss